

Module-3

Laplace Transform

Let $F(t)$ be a function of t defined for all $t \geq 0$, then the Laplace Transform of $F(t)$ is defined as $L(F(t))$, which is given by,

$$L(F(t)) = \int_0^{\infty} e^{-pt} F(t) dt = f(p)$$

Provided that the integral exist, p is a parameter which may be real or complex. The function $f(p)$ is called image of $F(t)$.

Laplace Transform of Elementary Functions:

1. $L(1) = \frac{1}{p}, p > 0$

2. $L(t) = \frac{1}{p^2}, p > 0$

3. $L(t^n) = \frac{n!}{p^{n+1}}, p > 0, n \text{ is a true integer}$

4. $L(t^n) = \frac{\sqrt{n+1}}{p^{n+1}}, p > 0, n > -1$

$$5. \quad L(e^{at}) = \frac{1}{p-a}, \quad p > a$$

$$6. \quad L(e^{-at}) = \frac{1}{p+a}$$

$$7. \quad L(\sin at) = \frac{a}{p^2 + a^2}, \quad p > 0$$

$$8. \quad L(\cos at) = \frac{p}{p^2 + a^2}, \quad p > 0$$

$$9. \quad L(\sinh at) = \frac{a}{p^2 - a^2}, \quad p > |a|$$

$$10. \quad L(\cosh at) = \frac{p}{p^2 - a^2}, \quad p > |a|$$

Linearity Property:

If C_1 & C_2 are constants and f & g are functions of t , then

$$L(C_1 f(t) + C_2 g(t)) = C_1 L(f(t)) + C_2 L(g(t))$$

Transformations of Discontinuous Functions:

1. First translation or first shifting property,

$$\text{If } L(F(t)) = f(p) \text{ then } L(e^{at} F(t)) = f(p-a).$$

2. Second translation property or Heaviside's shifting theorem,

$$\text{If } L(F(t)) = f(p) \text{ \& } G(t) = \begin{cases} F(t-a), & t \geq a \\ 0, & t < a \end{cases}$$

then

$$L(G(t)) = e^{-ap} f(p).$$

3. Change of scale property,

$$\text{If } L(F(t)) = f(p) \text{ then } L(F(at)) = \frac{1}{a} f\left(\frac{p}{a}\right).$$

Q.1) Find the Laplace Transform of,
 $7e^{2t} + 9e^{-2t} + 5 \cos t + 7t^3 + 5 \sin 3t + 2$.

Solⁿ:

$$\text{Let, } F(t) = 7e^{2t} + 9e^{-2t} + 5 \cos t + 7t^3 + 5 \sin 3t + 2$$

Now,

$$L(F(t)) = L(7e^{2t} + 9e^{-2t} + 5 \cos t + 7t^3 + 5 \sin 3t + 2)$$

$$= 7 L(e^{2t}) + 9 L(e^{-2t}) + 5 L(\cos t) + 7 L(t^3) + 5 L(\sin 3t) + 2 L(1)$$

$$= \frac{7}{p-2} + \frac{9}{p+2} + \frac{5p}{p^2+1} + \frac{7 \times 3!}{p^{3+1}} + \frac{5 \times 3}{p^2+9} + \frac{2}{p}$$

$$= \frac{7}{p-2} + \frac{9}{p+2} + \frac{5p}{p^2+1} + \frac{42}{p^4} + \frac{15}{p^2+9} + \frac{2}{p}$$

Q.2) Find Laplace transform of (i) $\sin 2t \cdot \cos 3t$
(ii) $\sin^3 2t$

Solⁿ:

(i) $F(t) = \sin 2t \cdot \cos 3t$

$$\Rightarrow F(t) = \frac{1}{2} [2 \sin 2t \cdot \cos 3t]$$

$$= \frac{1}{2} [\sin (2t+3t) + \sin (2t-3t)]$$

$$= \frac{1}{2} [\sin 5t + \sin (-t)]$$

$$\Rightarrow F(t) = \frac{1}{2} (\sin 5t - \sin t)$$

Now,

$$L(F(t)) = \frac{1}{2} [L(\sin 5t) - L(\sin t)]$$

$$= \frac{1}{2} \left[\frac{5}{p^2+25} - \frac{1}{p^2+1} \right]$$

$$\therefore L(F(t)) = \frac{5}{2(p^2+25)} - \frac{1}{2(p^2+1)}$$

(ii) $F(t) = \sin^3 2t$

We have, $\sin 3t = 3 \sin t - 4 \sin^3 t$

$$\Rightarrow \sin 6t = 3 \sin 2t - 4 \sin^3 2t$$

$$\Rightarrow \sin^3 2t = \frac{3}{4} \sin 2t - \frac{1}{4} \sin 6t$$

Now,

$$L(F(t)) = L(\sin^3 2t) = \frac{3}{4} L(\sin 2t) - \frac{1}{4} L(\sin 6t)$$

$$= \frac{3}{\cancel{2}4} \left[\frac{\cancel{2}}{p^2+4} \right] - \frac{1}{\cancel{4}^2} \left[\frac{\cancel{6}^3}{p^2+36} \right]$$

$$= \frac{3}{2(p^2+4)} - \frac{3}{2(p^2+36)}$$

$$= \frac{3}{2} \left[\frac{\cancel{p^2+36} - \cancel{p^2} - 4}{(p^2+4)(p^2+36)} \right]$$

$$= \frac{3 \times \cancel{32}^{16}}{2(p^2+4)(p^2+36)}$$

$$\therefore L(F(t)) = \frac{48}{(p^2+4)(p^2+36)}$$

Q.3) Find the Laplace transform of $e^t t^{-1/2}$.

Solⁿ:

~~Let, $F(t) = e^t t^{-1/2}$~~

First translation property,

If $L(F(t)) = f(p)$ then

$$L(e^{at} F(t)) = f(p-a)$$

For $e^t t^{-1/2}$, $F(t) = t^{-1/2}$

$$f(p) = L(F(t)) = L(t^{-1/2}) = \frac{\sqrt{(-1/2)+1}}{p^{-1/2+1}} = \frac{\sqrt{1/2}}{p^{1/2}} = \frac{\sqrt{\pi}}{\sqrt{p}}$$

$$f(p) = \sqrt{\frac{\pi}{p}}$$

$$\Rightarrow f(p-a) = f(p-1) = \sqrt{\frac{\pi}{p-1}}$$

$$\therefore L(e^t t^{-1/2}) = \sqrt{\frac{\pi}{p-1}}$$

$$\int u v dx = u \int v dx - \int (u' \int v dx) dx$$

Q.4) Find the Laplace transform of

$$F(t) = \begin{cases} 1 & , 0 \leq t < 1 \\ +t & , 1 \leq t < 2 \\ t^2 & , 2 \leq t < \infty \end{cases}$$

Solⁿ:

$$L(F(t)) = \int_0^{\infty} e^{-pt} F(t) dt$$

$$= \int_0^1 e^{-pt} \cdot 1 dt + \int_1^2 e^{-pt} \cdot t dt + \int_2^{\infty} e^{-pt} t^2 dt$$

$$= \left[\frac{e^{-pt}}{-p} \right]_0^1 + \left[t \int e^{-pt} dt - \int (1 \int e^{-pt} dt) dt \right]_1^2$$

$$+ \left[t^2 \int e^{-pt} dt - \int (2t \int e^{-pt} dt) dt \right]_2^{\infty}$$

$$= \left[\frac{e^{-pt}}{-p} \right]_0^1 + \left[\frac{t e^{-pt}}{-p} - \frac{e^{-pt}}{p^2} \right]_1^2 + \left[\frac{t^2 e^{-pt}}{-p} - 2 \int \frac{t e^{-pt}}{-p} dt \right]_2^{\infty}$$

$$= \left[\frac{e^{-pt}}{-p} \right]_0^1 + \left[\frac{t e^{-pt}}{-p} - \frac{e^{-pt}}{p^2} \right]_1^2 + \left[\frac{t^2 e^{-pt}}{-p} + \frac{2}{p} \int t e^{-pt} dt \right]_2^{\infty}$$

$$= \left[\frac{e^{-pt}}{-p} \right]_0^1 + \left[\frac{t e^{-pt}}{-p} - \frac{e^{-pt}}{p^2} \right]_1^2 + \left[\frac{t^2 e^{-pt}}{-p} + \frac{2}{p} \left[\frac{t e^{-pt}}{-p} - \frac{e^{-pt}}{p^2} \right] \right]_2^{\infty}$$

$$\boxed{e^{-\infty} = 0}$$

$$= -\frac{1}{p} [e^{-p} - 1] + \left[\frac{2e^{-2p}}{-p} - \frac{e^{-2p}}{p^2} + \frac{e^{-p}}{p} + \frac{e^{-p}}{p^2} \right]$$

$$+ \left[\frac{4e^{-\infty}}{p} - \frac{4e^{-2p}}{p^2} - \frac{2e^{-2p}}{p^3} \right]$$

$$= \cancel{-\frac{e^{-p}}{p}} + \frac{1}{p} - \frac{2e^{-2p}}{p} - \frac{e^{-2p}}{p^2} + \cancel{\frac{e^{-p}}{p}} + \frac{e^{-p}}{p^2} + \frac{4e^{-2p}}{p}$$

$$+ \frac{4e^{-2p}}{p^2} + \frac{2e^{-2p}}{p^3}$$

$$\therefore L(F(t)) = \frac{1}{p} + \frac{2e^{-2p}}{p} + \frac{3e^{-2p}}{p^2} + \frac{e^{-p}}{p^2} + \frac{2e^{-2p}}{p^3}$$

Q.5) Given that $L(\cos^2 t) = \frac{p^2+2}{p(p^2+4)}$. Find $L(\cos^2 at)$.

Solⁿ:

$$\text{Let, } F(t) = \cos^2 t$$

$$\Rightarrow F(at) = \cos^2 at$$

$$L(F(at)) = \frac{1}{a} f\left(\frac{p}{a}\right)$$

$$f(p) = L(F(t))$$

$$\Rightarrow f(p) = \frac{p^2+2}{p(p^2+4)}$$

$$\Rightarrow f\left(\frac{p}{a}\right) = \frac{(p/a)^2+2}{(p/a)(p^2/a^2+4)} = \frac{\cancel{a^3}^a(p^2+2a^2)}{\cancel{a^2}^1[p(p^2+4a^2)]}$$

$$\therefore L(\cos^2 at) = \frac{1}{a} \left[\frac{a(p^2+2a^2)}{p(p^2+4a^2)} \right] = \frac{p^2+2a^2}{p(p^2+4a^2)}.$$

Functions of Exponential Order

A function $F(t)$ is said to be of exponential order as $t \rightarrow \infty$ if there exist a constant M and b , and a fixed value t_0 & t such that,

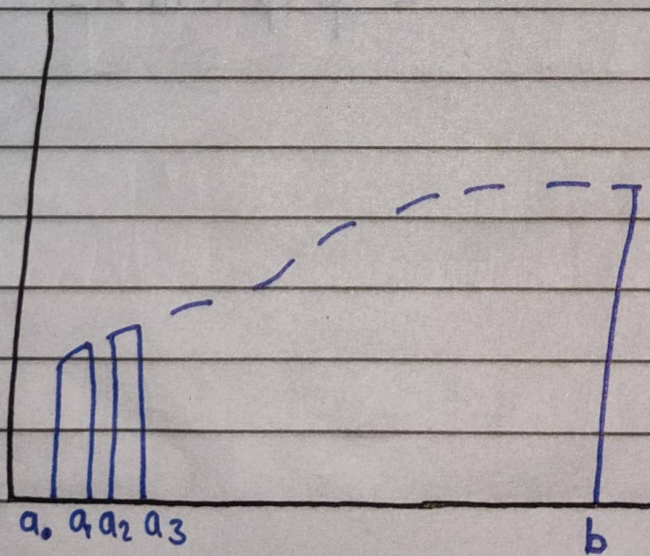
$$|F(t)| < M e^{bt}, \quad t \geq t_0, \quad t \rightarrow \infty$$

We also write,

$F(t) = O(e^{bt})$, $t \rightarrow \infty$ to mean that $F(t)$ is of Exponential order.

Function of class 'A'

A function which is sectionally or piece-wise continuous over every finite interval in the range $t \geq 0$ and is of exponential order as $t \rightarrow \infty$ is termed as a function of class 'A'.



Existence Theorem

If a function $F(t)$ is of class 'A' then $L(F(t))$ exists.

Q.1) Prove that t^n is a function of exponential order as $t \rightarrow \infty$.

Solⁿ:

$$\text{Let, } F(t) = t^n$$

$$\lim_{t \rightarrow \infty} e^{-bt} \cdot t^n = \lim_{t \rightarrow \infty} \frac{t^n}{e^{bt}} = 0 \quad \left[\begin{array}{l} e^\infty = \infty, 1 = 0 \\ \infty \end{array} \right] \text{ (exists)}$$

$$\left[e^{-bt} F(t) < M \right]$$

So, t^n is a function of Exponential Order.

Q.2) Show that e^{t^2} is not of exponential order as $t \rightarrow \infty$.

Solⁿ:

$$\lim_{t \rightarrow \infty} e^{-bt} \cdot e^{t^2} = \lim_{t \rightarrow \infty} e^{t^2 - bt} = \lim_{t \rightarrow \infty} e^{t(t-b)}$$

If $t = b$,

$$\lim_{t \rightarrow \infty} e^0 = 1$$

If $b < t$,

$$\lim_{t \rightarrow \infty} e^{t(t-b)} = \infty$$

If $b > t$,

$$\lim_{t \rightarrow \infty} e^{t(t-b)} = 0$$

$\therefore$ For $b < t$,

$$\lim_{t \rightarrow \infty} e^{-bt} \cdot e^{t^2} \text{ does not exist.}$$

$\therefore$ It is not of exponential order.

#① Laplace Transform of Derivative of Function.

$$L(F'(t)) = p L(F(t)) - F(0) = p f(p) - f(0)$$

② Laplace Transform of Derivatives of n^{th} order.

$$L(F^n(t)) = p^n L(F(t)) - p^{n-1} F(0) - p^{n-2} F'(0) - p^{n-3} F''(0) \dots F^{n-1}(0).$$

③ Laplace of Function Multiplied by t^n .

If $L(F(t)) = f(p)$ then

$$L(t^n F(t)) = (-1)^n \frac{d^n}{dp^n} f(p) = (-1)^n \frac{d^n}{dp^n} L(F(t))$$

④ If $L(F(t)) = f(p)$ then

$$L\left(\int_0^t F(t) dt\right) = \frac{f(p)}{p}$$

⑤ Division by t .

If $L(F(t)) = f(p)$ then

$$L\left(\frac{F(t)}{t}\right) = \int_p^\infty f(p) dp.$$

Q.1) If $F(t) = \frac{e^{at} - \cos bt}{t}$, find $L(F(t))$.

Solⁿ:

$$\text{Let, } F(t) = \frac{G(t)}{t} = \frac{e^{at} - \cos bt}{t}$$

$$G(t) = e^{at} - \cos bt$$

$$L(F(t)) = L\left(\frac{G(t)}{t}\right) = \int_p^\infty g(p) dp$$

$$[g(p) = L(G(t))]$$

Now,

$$g(p) = L(G(t)) = L(e^{at} - \cos bt)$$

$$\Rightarrow g(p) = L(e^{at}) - L(\cos bt)$$

$$\Rightarrow g(p) = \frac{1}{p-a} - \frac{p}{p^2+b^2}$$

$$L(F(t)) = L\left(\frac{G(t)}{t}\right) = \int_p^\infty \left[\frac{1}{p-a} - \frac{p}{p^2+b^2} \right] dp$$

$$= \int_p^\infty \frac{dp}{(p-a)} - \int_p^\infty \frac{p}{(p^2+b^2)} dp$$

$$= \left[\log(p-a) \right]_p^\infty - \frac{1}{2} \int_{(p^2+b^2)}^\infty \frac{dx}{x}$$

$$\left[\begin{array}{l} \text{let } x = p^2 + b^2 \\ \Rightarrow dx = 2p dp \end{array} \right]$$

$$= \left[\log (p-a) \right]_p^{\infty} - \frac{1}{2} \left[\log x \right]_{(p^2+b^2)}^{\infty}$$

$$= \left[\log (p-a) \right]_p^{\infty} - \frac{1}{2} \left[\log (p^2+b^2) \right]_p^{\infty}$$

$$= \left[\log (p-a) - \log \sqrt{p^2+b^2} \right]_p^{\infty}$$

$$= \left[\log \left(\frac{p-a}{\sqrt{p^2+b^2}} \right) \right]_p^{\infty}$$

$$= \left[\log \left(\frac{p(1-a/p)}{p\sqrt{1+b^2/p^2}} \right) \right]_p^{\infty}$$

$$= \log \left(\frac{1-a/\infty}{\sqrt{1+b^2/\infty}} \right) - \log \left(\frac{1-a/p}{\sqrt{1+b^2/p^2}} \right)$$

$$= \log(1) \overset{0}{\rightarrow} - \log \left(\frac{p-a}{\sqrt{p^2+b^2}} \right)$$

$$= - \log \left(\frac{p-a}{\sqrt{p^2+b^2}} \right)$$

$$= \log \left(\frac{\sqrt{p^2+b^2}}{p-a} \right).$$